SIMULTANEOUS *Go* VIA OBJECTIVE QUANTUM REDUCTION

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Abstract. We construct a simultaneous combinatorial game *SGo* that is in a precise mathematical sense a simple symmetric simultaneization of the classical board game *Go*, where simplicity too is a precise mathematical notion. We develop some general theory of this symmetric simultaneization here, in particular on the universal level, and this may be of independent interest. Based on this we prove a theorem that chess and *Go* are qualitatively different in this theory. *SGo* is not just theoretical: it is a very playable game that is in some ways a simpler game than *Go*, as Komi, Ko and suicide rules are removed. On the other hand it has similar and ostensibly sharper dynamics, a fact we verify via computer simulation. Using the argument of Nash, we show that a finite symmetric simultaneous combinatorial game has a mixed equilibrium strategy that draws on average, and so *SGo* does as well. The construction of *SGo* is inspired by physical quantum theory, despite the game being combinatorial.

1. Introduction

Go is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. However, this structure also has drawbacks: we need an ad hoc rule such as Komi to deal with the advantage of Black having the first turn. Moreover, the Ko situation and its cousins like triple Ko necessitate certain non repetition rules. Although non repetition rules are necessary in the end game of many types of games, like chess for example, in *Go* they are necessary even in the early game, which is to say the least inelegant. Furthermore, this leads to Ko fights, in which the game loses its local-to-global character[1].

At the same time, partly due to its local-to-global nature, *Go* is almost simultaneous. We make this precise by proving that *Go* has a simple (in a certain mathematical sense) symmetric simultaneization, see Theorem 5.2. The construction of a particular such simultaneization, called *SGo*, is the main thrust of this note. We also construct a theoretical variant of it, *DSGo*, which in addition has a faithfulness property relative to *Go*.[2] In contrast, we show that this fails for chess: suitably formalized chess has no simple simultaneization that stays faithful to classical play, see Theorem 5.7. *SGo* is a combinatorial symmetric simultaneous game, which is playable on a standard *Go* board, although we need a third type of stone that we call q-stones.[3] It is in some ways a simpler game than *Go* as it

[1] In the sense that if a player engages in a Ko fight, they must make moves that are strategically disconnected.

[2] D stands for decoherence.

[3] Needed in the following formal sense: a position of *SGo* carries a set of classical branches, which the displayed board summarizes, see Section 2.1; an intersection on which the branches disagree can be displayed neither black nor white, forcing a third marker. That the board with

loses the Komi, Ko and suicide rules, and it has just two rules: stone placement and capture (aside from the end of game rule). Its game dynamics are very similar to the classical *Go*; we make this precise via some empirical studies outlined in Section 2.2. For example, on the 19×19 board, random games of *SGo* appear to be at least as sharp as those of *Go* (in fact the expected absolute game margin is much higher, but this must be properly interpreted). Furthermore, random *SGo* games at least empirically self-terminate (398 mean turns at 19×19, with no non repetition rules invoked, in every one of a thousand games); in contrast, half of the *Go* games were still running after 1500 turns.[4]

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal simultaneization of certain sequential games developed in Section 4.3, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics, cf. Section 2.2. To get interesting dynamics in the case of *SGo*, we also have "objective reduction of the quantum state" – it is a deterministic, partial collapse of branches based on objective board information. The term objective reduction is coined by Penrose, see for instance [15]. Accordingly, a position of *SGo* carries its set of branches with it, as a position of chess carries its castling rights, see Section 2. Of course, the word quantum is used figuratively: there is no unitary time evolution, no complex amplitudes, and no probabilistic outcomes. The game is combinatorial and deterministic, and it would not be appropriate to call it "quantum Go". For games with genuine quantum features see for instance [6], and for a quantum version of *Go* itself [17].

Using the famous idea of Nash in [13], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said, finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw, not just to draw on average; on the 2×2 board both players do, but already on the 3×3 board neither player has one, so optimal play is necessarily mixed (Section 4.1).

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [2], [19]. It also led to Conway's theory of surreal numbers [4]. There is also research on simultaneous and synchronized combinatorial games, see for instance [3], [9], [10]. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

2. Game rules

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other's move on this turn. We allow pass moves as in *Go*. Simultaneous action requires

this single extra stone type suffices to referee the game is the simplicity assertion of Theorem 5.2, cf. Remark 4.21.

[4]These results come with caveats, a major one being that our rules for both games allow suicide, which is not standard. Furthermore, informed play is of course expected to produce vastly shorter games, for *Go* especially.

modifying the placement and capture rules of *Go*, while removing the Komi, Ko, and suicide rules. We also need to introduce a third type of stone, called ***q-stones*** (quantum stones), which may be thought of as simultaneously white and black. On the board a q-stone is displayed *red* or, as a visual aid, *blue* when a capture is currently blocked through it, in the sense of the capture conditions below; the colors red and blue are purely visual and carry no rule content.

2.1. **The entanglement state.** As with chess, whose position includes the castling and en passant rights in addition to the piece diagram, a position of *SGo* carries information beyond the stone diagram: its ***entanglement state***. (As with those rights, it can of course be recovered from the recorded game history.) This is a finite, nonempty set of classical *Go* positions, called ***branches***, representing the classical positions that the play so far may have produced. On every turn each branch evolves as a classical *Go* position: the two moves of the turn are played on it in both orders, giving up to two new branches per branch, where a move whose intersection is occupied is forfeited (the player effectively passes in that branch). The capture resolution rules below then prune branches, using only objective visible board information as triggers.

The board that the players see and play on is maintained by the rules below alone. It is a ***summary*** of the game history. Its role in the semantics is to supply the objective information by which the entanglement is reduced: branches incompatible with the displayed board are discarded. The summary may be read as a *projection* of the entanglement state. An occupied intersection is occupied in some branch. A displayed black or white stone has that color in some branch. A q-stone is black in some branch and white in another: a stone that is simultaneously white and black, over the branches. Near classical play this reading is nearly exact; in the decoherent variant of Section 5.1 it is exact. Already one turn from classical play the display and the branches can genuinely diverge, and the more so in deeply entangled positions. This is a quantum-like effect, and we return to it in Section 4.3. The role of the capture resolution rules below is to prune branches – to collapse entanglement – based on objective board information. In practice the entanglement state is small, and the play in Section 3 can be followed on the board alone. The formal treatment is given in Section 4.3, where Theorem 5.2 becomes essentially a matter of the definitions.

2.2. **Motivation for the rules.** Before we proceed to the main rules, we briefly overview the motivation and constraints we impose. We require:

(1) *SGo* forms a simple simultaneization of *Go*, in the precise sense of Theorem 5.2. Being a simultaneization loosely means that *SGo* naturally maps to the universal simultaneization of *Go*. The decoherent variant of Section 5.1 moreover keeps the displayed board faithful to the entanglement near classical play (Theorem 5.2); *SGo* itself approximates this, embracing the divergences.
(2) It is playable on a standard *Go* board, which in particular requires us to represent simultaneous moves to the same position, and we choose to do this by introducing q-stones (visually red or blue). In this case there is no real choice, as any two faithful ways to represent such simultaneous action are clearly equivalent. We are representing the branched state where

White moves to an intersection and Black passes, and Black moves to that intersection and White passes.

(3) We want the minimum amount of rules, in particular we do not want ad hoc rules dealing with Ko type situations.

(4) We want *SGo* to have similar game dynamics to the standard *Go*, that is with nearly optimal play the positions should tend to become very sharp. This is far from the case for the universal simultaneization $\mathrm{Sim}(Go)$ of *Go*, as considered in Section 4.3. For $\mathrm{Sim}(Go)$, games tend to be drawn due to players lacking forcing options. Although we cannot yet present the above as formal theorems, we will discuss just below some extensive empirical evidence based on computer simulation of *SGo* play.

To achieve these dynamics, we want to "optimally" prune branched histories resulting from simultaneous action, using objective global board information. This is formalized by the capture resolution rule. This has the effect of giving players very forcing options, see Section 3.2 for some examples. But this is not without constraints: we need to prune branches so that the game stays symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 2.4. We also cannot prune too much, for example in the suicidal capture condition in Definition 2.2 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 2.3.

Requirement (4) is supported by computer experiments.[5] The main comparison plays both games on their common rule base – the variant of *Go* fixed before Theorem 5.2 (suicide legal, no superko rule) – under uniformly random play over all legal board moves, passing only when none is available; 1000 games per variant (200 in the last column). On the 5×5 board the game bound of Section 4.1 is set to 40 turns; on 19×19, *SGo* is left unbounded, while *Go* is stopped at 1500 turns:

	SGo 5×5	*Go* 5×5	*SGo* 19×19	*Go* 19×19
mean margin	13.6	12.0	235.3	68.8
draws	4.6%	5.4%	0.0%	4.5%
games ending on their own	99%	0%	100%	50%
mean length in turns	22.4	39.9	398	1106
q-stones created per game	3.3	–	8.6	–
q-stones on the final board	2.3	–	5.0	–
peak number of branches	4.2	–	8.7	–
final number of branches	3.4	–	3.7	–

Here the mean margin is the absolute difference of the two area scores, and a game ends on its own when the end of the game rule fires before the bound or cap (the 19×19 figures for *Go* are censored at the cap, so its true mean length is larger still). On 5×5 the margins are nearly identical and the draw rates similar, and the difference is termination: 99% of *SGo* games ended of their own accord within the bound – and with the bound lifted, every one of them does, in 22.4 turns on average – while every classical game was still churning when the bound stopped it. On 19×19 the difference is dramatic on both counts: random *SGo* ends by itself

[5]The simulations and exact computations reported in this paper, together with extensive machine verification of its rules, examples, and arguments – including an exhaustive census of the one-turn faithfulness conditions over every legal classical position of the 3×3 board – were carried out by Claude, an AI system by Anthropic.

on a decisively settled board, with mean margin exceeding half the board, while half of the classical games were still cycling, far from settled, when stopped (every one of the thousand *SGo* games ended on its own).

Note that the margin is a roughly constant fraction of the board across sizes (55% at 5×5, 65% at 19×19), rather than the $\sqrt{\text{area}}$ scaling that would hold if the stone settlements had low global correlations.[6] A random *SGo* game settles through global capture avalanches, and the final board is macroscopically one sided. The much smaller margin in the case of *Go* is likely partly due to the fact that at the game cap many games were far from settled, so that the comparison should not be oversold.

Disabling the capture resolution – approximating the dynamics of the universal simultaneization of Section 4.3 – collapses the game completely: no stone is then ever removed, the two colors necessarily place equal numbers of stones, and all 500 simulated 5×5 games ended in exact draws. The capture resolution is the entire engine of the dynamics; the placement rule is dictated by the entanglement semantics.

We do not claim that the rules below are uniquely determined by the above requirements; what we show is that the requirements can be met at all, which by Theorem 5.7 is not automatic. Necessity is discussed rule by rule: Remark 2.3 describes what fails without the corresponding clause, and the disabling experiment above shows the role of the capture resolution as a whole.

One may wonder how informed play changes the above distributions; we partially address this in Section 6.1.

Finally, the last four rows of the table show that the entanglement stays small: a finished *SGo* game displays on average about two q-stones on the small board, and five among the 361 points of the large one, so the displayed board stays nearly classical. The branch counts are obtained by tracking the branch sets consistent with the displayed boards – an upper bound for the entanglement state (all 1000 of the 19×19 games were so tracked). This is a separate statistic from the q-stone count: branches may disagree in occupancy alone, which the display shows as an ordinary black or white stone with no q-stone. At both sizes the final branch count stays of order one, essentially independent of board size (3.4 against 3.7) – a handful of branches on boards whose a priori branching is astronomical. For sharpness under optimal play, see the discussion of drawing strategies in Section 4.1.

2.3. **Some definitions.** Given a board state S, that is any (at the moment no legality restrictions) collection of white, black, q-stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A ***unicolor component*** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C ***black***, respectively

[6] We leave this vague, but we are alluding to the central limit theorem, which would apply if the local settlements were nearly independent.

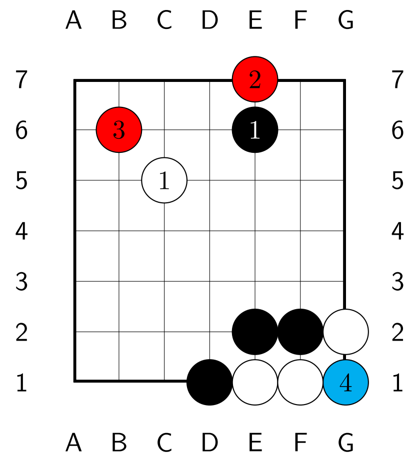
white. We may less often refer to ***q-components***, analogously defined as connected components of q-stones.

As in classical *Go*, a connected component C ***has no liberties*** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a ***trapped component***.

We define stones ***adjacent*** to C as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

2.4. **Stone placement rule of *SGo*.** If the players move to the same intersection i on the board, a q-stone is placed at i.[7]

To the right is an example of the placement rule. On move one the players move to different intersections and both stones are placed as in classical *Go*. On moves two, three and four they move to the same intersection – E7, B6 and G1 respectively – and each conflict yields a q-stone. (One can check that no capture then occurs on turn 4: the blue stone blocks both parts of condition 3 for the white component E1, F1.)



2.5. **Capture rule of *SGo*.** After both White and Black move on the current turn numbered n, using the stone placement rule above we get some board state S.

Definition 2.1. We say that a black, white, or q-component C is ***trapped on turn*** n if C is trapped and n is the last turn on which a stone was placed in or adjacent to C. A stone is called ***trapped*** on turn n, if it is an element of C as above. We say that a q-stone is ***trapped by white respectively black on turn*** n (white, respectively black, being the capturing side), if replacing it by a black respectively white stone it is trapped on turn n. For a black respectively white stone, trapped by white respectively black on turn n will simply mean trapped on turn n.

Definition 2.2. A white component C is resolved as ***captured on this turn*** if the following is satisfied:

(1) C is trapped on this turn.
(2) If C is adjacent to a q-stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 2.4 does not work, and the game evolution will be ill defined.)
(3) One of the following holds:

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[7] In terms of the entanglement state: each branch gains the stone of one of the two players (in each serialization the second move to i is forfeited), and all branches are kept. The branches then disagree at i, so displaying either black or white would misrepresent half of the entanglement: a q-stone is the faithful summary, and the rule is forced.

(a) No black stone or q-stone adjacent to C is trapped by white on this turn.
(b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then the component of that stone contains a stone that is created on this turn and is not a q-stone.

Remark 2.3. *The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.2. For example, if we remove the part of the condition "is not a q-stone", then Ko type infinite loops will become possible later. If we remove the condition "no stone in C is created on this turn", then Theorem 5.2 will fail.*

In the case C is captured:

- The stones of C are removed as prisoners of Black as in *Go*.
- Any q-stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

A stone converted this way is not considered to be placed or created on this turn; it retains the placement turn of the q-stone it replaces.

If C is a black component the process is naturally mirrored. As far as the rules are concerned this process is simultaneous for all captured components, cf. Lemma 2.4.

2.6. **Stage one reduction.** The above capture procedure is called ***stage one reduction*** and the resulting board state will be denoted by $R_n(S)$.

Lemma 2.4. *The stage one reduction is well defined. Namely, no q-stone is adjacent to components of both colors captured on this turn, so that the displayed conversion of q-stones is unambiguous; and the prunings of the entanglement state are unambiguous set operations.*

Proof. Suppose that a q-stone r is adjacent to unicolor components C_0, C_1, both captured on this turn, with C_0 white and C_1 black. As C_1 is captured, it is in particular trapped. So C_0 is adjacent to the q-stone r, which is itself adjacent to the trapped component C_1 of the opposite color, and Condition 2 fails for C_0. This contradicts the assumption that C_0 is captured.

Thus, each q-stone adjacent to some component captured on this turn is adjacent to captured components of only one color, and so its conversion is unambiguous. As the removal of stones is likewise unambiguous, each simultaneous resolution step, and hence the stage one reduction, is well defined. □

If the stage one reduction removes no stones, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction, denoted by $R^k(S)$, satisfies:

$$R^1(S) = R_n(S).$$
$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

That is, at stage k the capture conditions of Definition 2.2 are applied with "this turn" read as turn $n-k+1$, on the current board state: a capture of an earlier turn, blocked at that turn, may complete at a later one, once the blocking entanglement has collapsed. The recursion stops as soon as a stage removes no stones, and the turn is then resolved.

Remark 2.5. *In practice the resolution of most turns is easy to work out mentally; we will give some examples of "nested" entangled states further ahead. Of course having a computer work out turn resolution automatically is very helpful.*

2.7. **End of the game.** An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some black/white stones on the board as prisoners for White/Black. (As is common in standard play of *Go*.) Each black or white prisoner gives one point to the player who holds it. This is the main way of ending the game.
- The board position is unchanged after each of two consecutive turns. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row. (This is meant to ensure that a winning player can actually force a win when the opponent is stalling.)
- The board has no empty intersections.

We then decide the winner using Chinese *Go*-style rules (this is for simplicity). So the winner is decided by the integer sum of the prisoner count and area count. We now explain the latter.

We say that an empty intersection i on the board ***reaches black respectively white*** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is ***surrounded by black*** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a q-stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

(1) The number of black stones on the board.
(2) The number of empty intersections surrounded by black.

See Section 3.4 for an example.

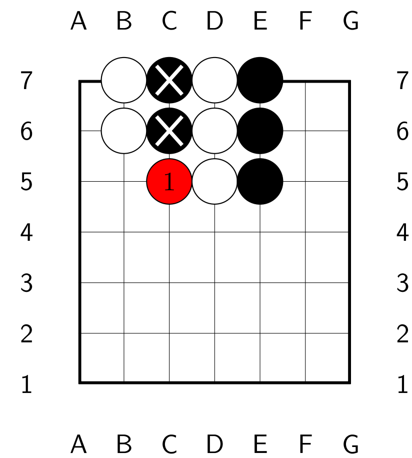
A draw is now possible, although compared to chess it is unlikely (for random games on the 19×19 board especially so, as the table in Section 2.2 shows).

3. Examples

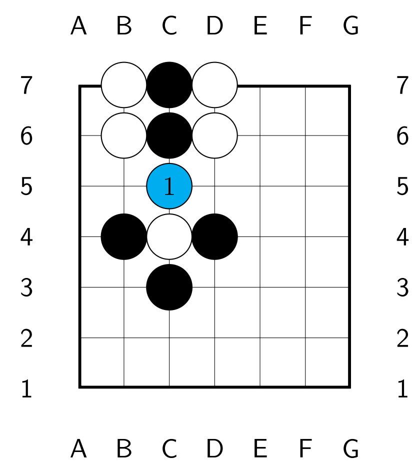
Throughout, a turn's joint move is written m/m', the move of the first player listed first – in *SGo*, Black's.

3.1. **Capture and non-capture moves.** Here are some examples and non-examples of capture.

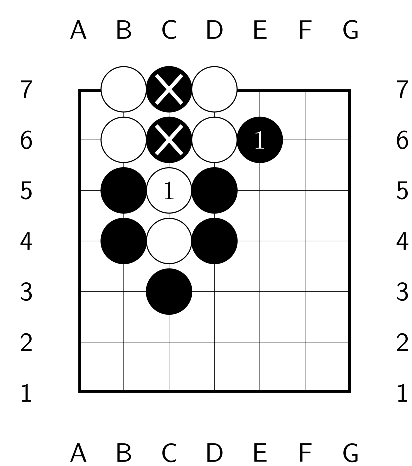
Turn 1 is $C5/C5$. This is a capture move, and C5 will resolve to white at the end of the turn.



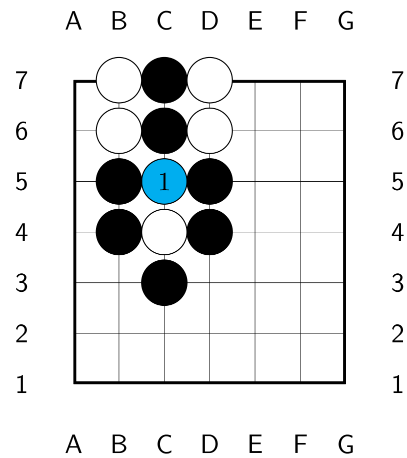
The turn is again $C5/C5$, in another position: not a capture move, as the created q-stone, marked blue, is adjacent to an opposite-colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



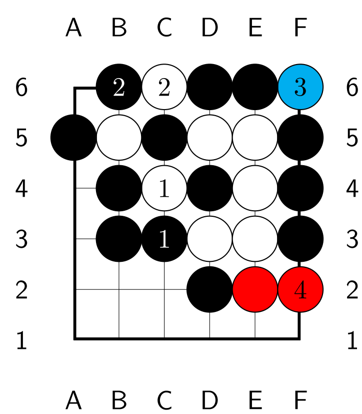
Turn 1 is $E6/C5$. White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture. Formally, this is forced by the condition that *SGo* is a simultaneization of *Go*, Theorem 5.2.



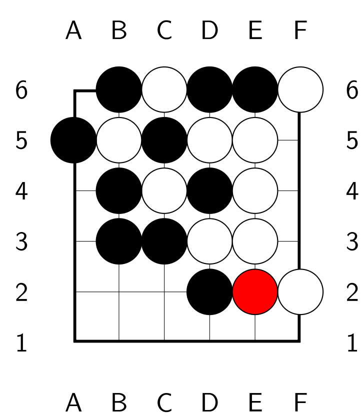
The turn is $C5/C5$: not a capture move, since Condition 3b is now not satisfied, the stone placed on this turn being a q-stone (marked blue, as the blocker).



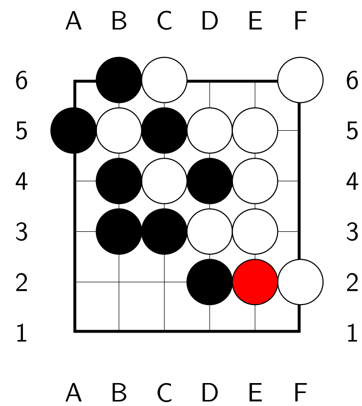
3.2. **More complex examples.**
In the position on the right, turn 1 is $C3/C4$ – Black attempts capture at C3, White at C4. Nothing is captured on turn 1 according to our rules, and after turn 2 the position becomes an example of a "nested entangled state". The continuation $B6/C6$; $F6/F6$; $F2/F2$ is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section; F6 is marked blue, being what blocks the capture of D6 and E6 on this turn. But F2 is a capture move for White since condition 3a is satisfied.



At the end of turn 4, stage one resolution is depicted on the right.

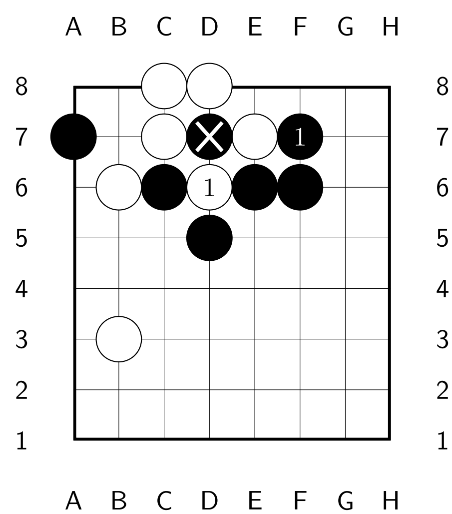


Stage two resolution. Stage three, which applies the capture conditions of turn 2, removes nothing: C5 and B5 each block the other's capture. So the recursion stops, and this is the final resolution of turn 4. Note that B5, C5, C4 and D4 are all trapped yet uncaptured – an entangled analogue of seki, impossible in classical *Go*: each capture is blocked by the entanglement of the others.

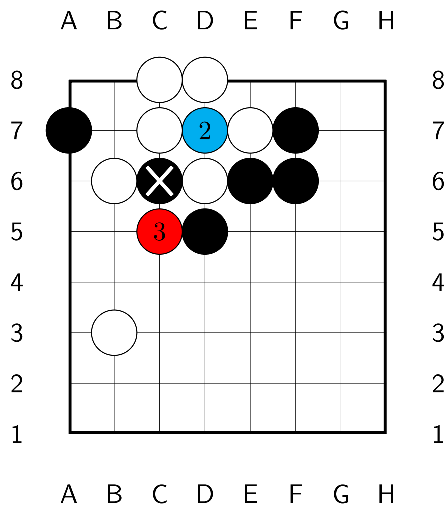


3.2.1. *Ko situation.* Below is a Ko type situation. Recall that we do not have a Ko rule; none is needed, since the board state cannot immediately loop by recapture.

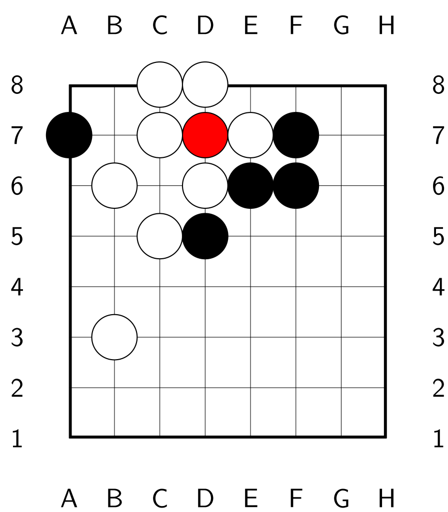
Turn 1 is $F7/D6$; White's D6 is a capture move.



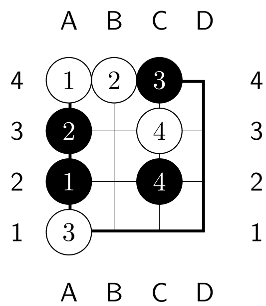
Here is the possible continuation $D7/D7$; $C5/C5$; the blue stone at D7 is what blocks Black's recapture of White's D6.

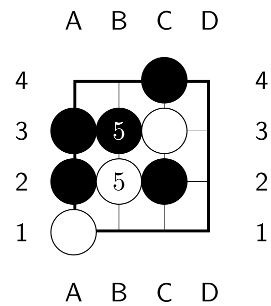


At the end of turn 3 we get the following position: the q-stone at D7 simply persists, and the Ko fight remains open, held in the entanglement.



Finally, here is an example where the component clause of the suicidal capture condition 3b – that the trapped neighbor's *component* contain a stone created on this turn – does the work. On the 4×4 board play the conflict-free history $A2/A4$; $A3/B4$; $C4/A1$; $C2/C3$:

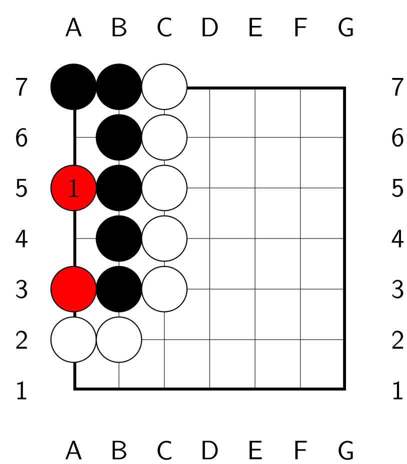




Now play $B3/B2$. Black's B3 fills the last liberty of the old white pair A4, B4, while White's B2 fills the last liberty of the black group A2, A3, B3. The black group is trapped by White on this turn, so condition 3a fails for the pair and the suicidal capture clause governs. The pair touches the trapped group at A3 – an old stone – but the group *contains* this turn's B3: the condition is met, and the pair is captured, as the right board shows. This is what the classical semantics demands: the two serializations of the turn agree, and each captures the pair – B3 played first captures it at once, and B2 then lands with the liberty B1; B2 played first, B3 still captures it, the black group surviving on the freed liberties. Had the clause read the adjacent stone alone, A3 being old, the capture would be refused, and the display would freeze on the board a pair that classical play removes in either order. The history is conflict-free, so the entanglement is a singleton throughout: the component clause is exactly what keeps the display classical here.

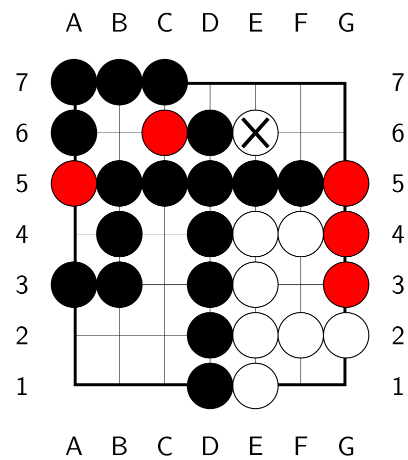
3.3. **Life and death.** Life and death is positionally very similar to *Go*; here is an example.

White cannot kill the black group: A4 and A6 are two eyes – a white stone placed in either is immediately captured, while a simultaneous conflict there yields a q-stone. In *SGo* the group is alive outright: even with Black's collaboration – Black filling one eye himself, White taking the other – the standing entanglement at A3 and A5 blocks every capture, and the group survives, wrapped in the entanglement.



3.4. **End game position.** In the diagram below, the players have agreed to end the game. The white stone at E6 is agreed to be a prisoner of Black. Assuming there are no other prisoners, White has 7+3 points and Black has $7{\cdot}7-(7+3)-5+1$

points, where the -5 accounts for the five q-stones, which count for neither player, and the $+1$ for the prisoner.



4. *SGo* AS A SYMMETRIC SIMULTANEOUS COMBINATORIAL GAME

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games. Such formalisms are standard in several communities: games with simultaneous moves appear as the deterministic case of the stochastic games of Shapley [21], and as the concurrent game structures of computer science [1]. Simultaneous play of combinatorial games as such is studied by Huggan–Nowakowski–Ottaway [9], see also the references therein; there, however, the effect of a pair of simultaneous moves must be specified by each ruleset separately, no general procedure being available. The emphasis here, on simultaneizations of a given sequential game and on their simplicity, appears to be new, and in this setting the pseudo pass moves together with the operation # further below do furnish such a general procedure. The inverse passage, representing a sequential game within an intrinsically simultaneous formalism, is standard in general game playing, and is likewise effected by pass-like moves: there the idle player is required to play an explicit "noop" move, cf. [22].

Definition 4.1. A ***combinatorial simultaneous game*** (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the ***pseudo pass move*** (it has multi grading both 0 and 1).
- Game states decompose as
$$S(G) = B(G) \sqcup B(G) \times M(G),$$
where $B(G)$ is some set (for example it might be the set of board states).

- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \to S(G),$$

which has the properties:
  1. $E(s, m)$ is undefined for s in one of the final states.
  2. If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
  3. If s is not final, $E(s, 0)$ is defined.
  4. If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' is 0).
  5. Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined, then $E(s, m_0, m_1)$ is defined.
  6. For s, m_0, m_1 as in the previous property, if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$

We say that G is ***finite*** if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \; E(q_0, m_1, \ldots, m_k) \text{ is defined.}$$

Note that any game history of length $2k$ can be rewritten in the form $h = \{m_i^0, m_i^1\}_{i=1}^{i=k}$, with $m_i^j \in M_j(G)$.

It is our convention that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(q_0, m_1, \ldots, m_k)$, for some m_i. For future use set $f(h) = E(q_0, m_1, \ldots, m_n)$.

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4.2. We say that a combinatorial simultaneous game G is ***symmetric*** if the following conditions are satisfied.

1. There is a bijection $\mathcal{A} : M(G) \to M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
2. There is a bijection $\mathcal{R} : B(G) \to B(G)$, satisfying $\mathcal{R}^2 = id$ and $\mathcal{R}(q_0) = q_0$, which we extend to $\mathcal{R} : S(G) \to S(G)$ by $\mathcal{R}(s, m) = (\mathcal{R}(s), \mathcal{A}(m))$, s.t. for all $s \in S(G)$, all $m \in M(G)$, and for $\mathcal{A}$ as above we have:
   - $\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
   - $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Lemma 4.3. *SGo is an ssc game, that is a symmetric simultaneous combinatorial game.*

Proof. Set $G = SGo$, with the latter described in Section 2. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pseudo pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is the full data maintained by the rules of Section 2: the displayed board with its placement records, together with the entanglement state of Section 2.1 – a finite set of legally obtainable positions of *Go*. The initial state is the empty board with the entanglement $\{q_0\}$, for q_0 the empty position.

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are moves to empty intersections of the displayed board, then $E((s, m_0), m_1) \in B(G)$ is defined to be the state produced by the resolution rules of Section 2: the displayed board resolves by the placement and capture rules (in *DSGo*, also the decoherence rule of Section 5.1), each branch evolves under the serialized moves, and the branches incompatible with the resolved display are discarded. Everything in this section applies verbatim to both games.

To see that G is symmetric, let $\mathcal{R} : B(G) \to B(G)$ be the map that interchanges the black and white colors of the stones in every branch (on the displayed board this interchanges black and white and leaves q-stones invariant). And let $\mathcal{A} : M(G) \to M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \to M_1(G)$, which then induces $\mathcal{A}$. Then clearly the conditions of Definition 4.2 will be satisfied. $\square$

Remark 4.4. *We have not formalized $F(SGo)$: ending a game by agreement is not an automaton notion. This caveat is however minor, and is already present in classical Go.*

4.1. **Solvability.** A theorem of Zermelo [20] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly *SGo* cannot have a winning strategy: mirroring a winning strategy via the symmetry operators would yield one for the opponent, and both players cannot win. Furthermore, unless we put an upper bound on the length of a game, or on the number of position repetitions, an *SGo* game can be infinitely long.[8] So we will assume there is such a bound, which will make *SGo* a finite game.

One may ask whether *SGo* has a *strategy to draw*: a pure strategy guaranteeing at least a draw. Exact computation shows that on the 2×2 board it does – both players can force a draw with any game length bound between 6 and 10 – but already on the 3×3 board, with any bound between 6 and 8, neither player has such a strategy: a player following any fixed pure strategy can be made to lose. Optimal play is thus necessarily mixed from the 3×3 board on, and we expect the same on all larger boards – the mixed equilibrium of Theorem 4.5 below is not a formality.

Theorem 4.5. *For any assignment of payoffs to a win and a draw, a finite combinatorial simultaneous game G (Definition 4.1) has a Nash equilibrium.*

Proof. For a non final $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \,|\, E(s, m) \text{ is defined}\},$$

[8] This is also true for *Go* and chess.

which is non-empty as it contains the pseudo pass move 0.

A ***pure strategy*** of player P_i assigns to each non final $s \in B(G)$ a move in $M_i(s)$, so that the set of pure strategies is the finite set

$$\Sigma_i = \prod_s M_i(s),$$

the product over the non final elements of $B(G)$. The space of mixed strategies of player P_i is then:

$$\mathcal{P}_i = \Delta(\Sigma_i),$$

the space of probability distributions over Σ_i, which is a compact convex simplex in $\mathbb{R}^N$, for some very large N.

A pair of pure strategies determines a play of the game, and so a payoff, by finiteness of G.[9] The expected payoff function on $\mathcal{P}_0 \times \mathcal{P}_1$ is then bilinear, in particular continuous. Moreover, for a fixed strategy of the opponent, the expected payoff of P_i is a linear function on the simplex $\mathcal{P}_i$, so that the set of countering strategies of P_i is a non-empty, closed and convex face of $\mathcal{P}_i$. Consequently, the multi valued function ϕ, assigning to a joint strategy its set of countering joint strategies, has a closed graph, and the image by ϕ of any point is non-empty and convex.

Then as argued by Nash [13] using Kakutani's fixed point theorem [12] we get existence of an equilibrium point. □

Corollary 4.6. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 for a draw. By the proof of Theorem 4.5, we may view G as a zero sum matrix payoff game, with the rows and columns of the payoff matrix indexed by the pure strategies Σ_0, Σ_1. By the classical theory of matrix payoff games (see for instance [14]), all Nash equilibria have the same expected payoff for Black, namely the value v of the matrix game, and an equilibrium strategy of Black guarantees Black an expected payoff of at least v against every strategy of White. So it is enough to show that $v = 0$; that a symmetric zero sum game has value zero is classical, cf. [7], and for completeness we give the argument.

Suppose otherwise, and suppose without loss of generality that $v > 0$, so that for an equilibrium joint strategy

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

Black has expected payoff $v > 0$.

Let $\mathcal{R}, \mathcal{A}$ be the symmetry operators as in the proof of Lemma 4.3. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has expected payoff v, that is with Black expected payoff $-v < 0$. As the latter must again be v, we have a contradiction. □

4.2. **Symmetric combinatorial sequential games.** Combinatorial sequential games have a beautiful Conway recursive definition [4]. However, for our purpose, in particular for the universal simultaneization construction, it is convenient to interpret them in terms of finite automata.

Definition 4.7. A ***combinatorial sequential game*** (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

[9] Possibly the finiteness condition can be relaxed.

- A deterministic automaton G, whose set of states is a set $S(G)$ called the set of game states of G.
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the ***pseudo pass move*** (it has multi grading both 0 and 1).
- A $\mathbb{Z}_2$ graded set of game states $S(G) = States(G) = S_0(G) \sqcup S_1(G)$, we let $deg : S(G) \to \mathbb{Z}_2$ denote the associated function.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S_0(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:
$$E = E_G : S(G) \times M(G) \to S(G),$$
which has the properties:
  1. $E(s, m)$ is undefined for s in one of the final states.
  2. $E(s, m)$ is defined only if m has the same grading as s or if $m = 0$.
  3. If $E(s, m)$ is defined then $deg(E(s, m)) = deg(s) + 1$.
  4. If s is not a final state then $E(s, 0)$ is defined.
  5. If $E(s, 0) = E(s', 0)$ then $s = s'$.
  6. If $E(s, 0, 0)$ is defined then it is a final state: two consecutive pseudo pass moves end the game.

We recursively define:
$$E(s, m_1, \ldots, m_k) = E(E(s, m_1, \ldots, m_{k-1}), m_k).$$
As before it will be understood that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(s, m_1, \ldots, m_k)$, for some m_i.

Example 4.8. *Note that Go has a pseudo pass move, but chess does not; however, chess can be augmented with a pseudo pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules, and with the game ending in a tie after two consecutive passes. This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in Zugzwang they may just pass). We will call this* **augmented chess** *denoted by* $\widetilde{chess}$.

Definition 4.9. We say that a combinatorial sequential game G is ***symmetric*** if the following conditions are satisfied.

1. There is a bijection $\mathcal{A} : M(G) \to M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{A}^2 = id$, $\mathcal{A}(0) = 0$.
2. There is a bijection $\mathcal{R} : S(G) \to S(G)$, satisfying $\mathcal{R}^2 = id$ and taking $S_0(G)$ to $S_1(G)$ s.t. for all $s \in S(G)$, all $m \in M(G)$ and for $\mathcal{A}$ as above we have:
$$\mathcal{R} \circ E(\mathcal{R}(s), \mathcal{A}(m)) = E(s, m),$$
whenever both sides are defined. Furthermore, if one side is defined then so is the other.
3. $\mathcal{R}$ also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Example 4.10. *Of course $\widetilde{\text{chess}}$ and classical Go are examples of symmetric games. In the case of Go, $\mathcal{A}$ interchanges black and white moves and $\mathcal{R}$ likewise interchanges the color of the stones on the board and shifts the initiative (it is easy to see that this does result in a legally reachable position).*

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{A}$, $\mathcal{R}$ implicitly fixed).

4.3. **Universal simultaneization.** Let G be a sequential game (by our standing assumption with a pass move 0 and symmetric). Let $\{m_i\}$ be a sequence of game moves. If some $m_i = 0$ and $m_k \neq 0$ for some $k > i$, then we say that it is an ***interior pass move***. We say that a finite sequence $\{m_i\}$ of moves of G is ***pseudo legal*** if it becomes a game history after adding or removing some interior pass moves from the sequence. It is not hard to see that given a pseudo legal sequence h, there is a unique minimal length game history as above, and we denote it by $leg(h)$.

For $a \in S(G)$ if there exists an $a' \in S(G)$ s.t. $E_G(a', 0) = a$ we set $\overline{a} = a'$, otherwise set $\overline{a} = E_G(a, 0)$. Property 5 in Definition 4.7 ensures that it does not matter which a' is chosen when it exists. For $a \in S(G)$ and $m \in M(G)$ set

$$a\#m = \begin{cases} E_G(a,m), & \text{if this is defined} \\ E_G(\overline{a},m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Let $\sim$ denote the ***pass equivalence***: the equivalence relation on the states of G generated by $s \sim E(s, 0)$, for s and $E(s, 0)$ non final. We call its classes ***pass classes***. By Property 5 and Property 6, a pass class contains at most one state of each grading. For a non final $b \in S(G)$ we let $\widehat{b}$, the ***grading normalization*** of b, denote the grading 0 member of the pass class of b when it exists, and $\widehat{b} = E_G(b, 0)$ otherwise; so $\widehat{b} = b$ for $b \in S_0(G)$, and $\widehat{b}$ is pass equivalent to b outside of the degenerate second case, in which a pass is appended. It is convenient to regard final states as carrying both gradings, as with the pseudo pass move, and for final b we set $\widehat{b} = b$. Define $P(a, m_0, m_1) \in 2^{S_0(G)}$, for $a \in S_0(G)$, by the following prescription in decreasing priority, where each term is understood to be replaced by its grading normalization, and the expression is defined if and only if all the terms are defined.

- $P(a, m_0, m_1) := a\#m_0\#m_1 \cup a\#m_1\#m_0$ (note that if $m_0 = m_1 = 0$ then the two terms coincide, so that $P(a, 0, 0) = a\#0\#0$.)
- $P(a, m_0, m_1) := a\#m_0\#m_1 \cup a\#m_1$.
- $P(a, m_0, m_1) := a\#m_0 \cup a\#m_1\#m_0$.
- $P(a, m_0, m_1) := a\#m_0 \cup a\#m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S_0(G) \times M_0(G) \times M_1(G) \to 2^{S_0(G)},$$

which we call the ***serialization map***: it evaluates the turn in its available serializations – both orders of play when both are defined, the degenerate cases otherwise – and collects the outcomes.

Definition 4.11. Let G be a sequential game. Its ***universal simultaneization*** is the simultaneous game $\mathrm{Sim}(G) = G'$ satisfying the following.

(1) $B(G') = 2^{S_0(G)}$. (The elements s of $\mathfrak{s} \in B(G')$ will be called ***state branches***.) By the above, an element of $B(G')$ may equivalently be regarded as a finite collection of pass classes, represented by their grading 0 members.
(2) The initial state $q_0' = \{q_0\}$.
(3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
(4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$; then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$, s is not final and $s\#m_i$ is defined.
(5) For $\mathfrak{s}$ and m_i as just above:
$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} P(s, m_0, m_1),$$
(undefined if some $P(s, m_0, m_1)$ is undefined).
(6)
$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s}\, s \in F(G).$$

Note that $\mathrm{Sim}(G)$ is uniquely defined by the above conditions up to the partition of its final states into W_0, W_1, D; these will not play a role but we can specify them as follows:

- $$\mathfrak{s} \in W_0(G') \iff \exists s \in \mathfrak{s}\,(s \text{ is winning for } P_0) \wedge \nexists s \in \mathfrak{s}\,(s \text{ is winning for } P_1).$$
- Similarly,
$$\mathfrak{s} \in W_1(G') \iff \exists s \in \mathfrak{s}\,(s \text{ is winning for } P_1) \wedge \nexists s \in \mathfrak{s}\,(s \text{ is winning for } P_0).$$
- In all other cases, $\mathfrak{s} \in D(G')$.

Remark 4.12. *The passage from positions to sets of positions is a form of the classical subset construction of automata theory* [16]. *In game theoretic terms, a state of* $\mathrm{Sim}(G)$ *is a knowledge state: the set of classical positions consistent with what the players have observed. Analogous knowledge based constructions are used in the study of games of imperfect information, see for instance* [18], [5].

Lemma 4.13. $\mathrm{Sim}(G)$ *is symmetric.*

Proof. Set $G' = \mathrm{Sim}(G)$, and let $\mathcal{A}, \mathcal{R}$ be the symmetry operators of G. Set $\widetilde{\mathcal{A}} : M(G') \to M(G')$ to just be $\mathcal{A}$. For
$$\{s_0, \ldots, s_k\} = \mathfrak{s} \in B(G'),$$
set
$$\widetilde{\mathcal{R}}(\mathfrak{s}) = \{\mathcal{R}(s_0), \ldots, \mathcal{R}(s_k)\},$$
with each $\mathcal{R}(s_i)$ replaced by its grading normalization, $\mathcal{R}$ being grading reversing. Since $\mathcal{A}(0) = 0$, the map $\mathcal{R}$ commutes with the pass move and so preserves pass equivalence; with Property 6 it follows that $\widetilde{\mathcal{R}}^2 = id$ exactly, and that the computation below is unaffected by the normalization. For
$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$
set
$$\widetilde{\mathcal{R}}(\mathfrak{s}, m) = (\widetilde{\mathcal{R}}(\mathfrak{s}), \widetilde{\mathcal{A}}(m)).$$

The needed symmetry properties for the maps $\widetilde{\mathcal{A}}, \widetilde{\mathcal{R}}$ are then verified using the symmetry properties of $\mathcal{A}, \mathcal{R}$ as follows. It is easily seen that $\overline{\mathcal{R}(a)} = \mathcal{R}(\overline{a})$, hence

$$\mathcal{R}(a)\#\mathcal{A}(m) = \begin{cases} E_G(\mathcal{R}(a), \mathcal{A}(m)) = \mathcal{R} \circ E_G(a, m), & \text{if defined} \\ E_G(\mathcal{R}(\overline{a}), \mathcal{A}(m)) = \mathcal{R} \circ E_G(\overline{a}, m), & \text{if defined} \end{cases}$$

and so

$$\mathcal{R}(a)\#\mathcal{A}(m) = \mathcal{R}(a\#m), \tag{4.14}$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$; then we get:

$$\begin{aligned} \widetilde{\mathcal{R}}(E_{G'}(\widetilde{\mathcal{R}}(\mathfrak{s}), \widetilde{\mathcal{A}}(m_0), \widetilde{\mathcal{A}}(m_1))) &= \widetilde{\mathcal{R}} \circ \cup_{s\in\mathfrak{s}} P(\mathcal{R}(s), \mathcal{A}(m_0), \mathcal{A}(m_1)) \\ &= \widetilde{\mathcal{R}} \circ \cup_{s\in\mathfrak{s}} \mathcal{R} \circ P(s, m_0, m_1) \text{ using (4.14)} \\ &= \cup_{s\in\mathfrak{s}} P(s, m_0, m_1) \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies property 2 of Definition 4.2 for pair moves at states of $B(G')$; the rest is similar. □

Definition 4.15. Let G_0 be a sequential game and G_1 a symmetric simultaneous game with $M(G_1) = M(G_0)$. The ***semi-classical*** states of G_1, with their ***associated positions*** in G_0, are defined recursively: the initial state of G_1 is semi-classical, with associated position the initial state of G_0. If s is semi-classical with associated position a, and m_0, m_1 *commute at* a – both $a\#m_0\#m_1$ and $a\#m_1\#m_0$ are defined and pass equivalent – then $E_{G_1}(s, m_0, m_1)$, if defined, is semi-classical, with associated position the common grading normalization of $a\#m_0\#m_1$ and $a\#m_1\#m_0$.

Definition 4.16. Let G_0 be a sequential game. We say that G_1 is a ***simultaneization of*** G_0 if the following is satisfied.

1. G_1 is a symmetric simultaneous game.
2. $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
3. There is a map $\rho : B(G_1) \to B(\mathrm{Sim}(G_0))$ satisfying $\rho(q_0) = q_0'$ and s.t. for $s \in B(G_1)$: whenever the state $E_{G_1}(s, m_0, m_1)$ is defined, so is $E_{\mathrm{Sim}(G_0)}(\rho(s), m_0, m_1)$, and
$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\mathrm{Sim}(G_0)}(\rho(s), m_0, m_1).$$
4. For every semi-classical s with associated position a (Definition 4.15): $\rho(s) = \{a\}$.
5. If s is semi-classical and $\rho(E_{G_1}(s, m_0, m_1))$ is a singleton, then $a\#m_0\#m_1$ and $a\#m_1\#m_0$ are pass equivalent, for a the associated position of s.

For $a \in S(G_0)$ set

$$\mathbb{M}(a) = \{m \in M(G_0) \,|\, a\#m \text{ is defined}\},$$

which we call the ***interface*** of the position a: the set of moves of either player available at a, with the operation $\#$ absorbing the turn order. Likewise, for $s \in B(G_1)$ set $M(s) = \{m \in M(G_1) \,|\, E_{G_1}(s, m) \text{ is defined}\}$, the interface of the state s.

Lemma 4.17. *Let G_1 be a simultaneization of G_0. Then for every non final $s \in B(G_1)$: every branch of $\rho(s)$ is non final, and*

$$M(s) \subseteq \bigcap_{a \in \rho(s)} \mathbb{M}(a).$$

That is, a move available at s is available in every branch. Equality of the two sides, together with $\rho(s) \neq \emptyset$, is precisely the statement that a joint move is available at s if and only if it is available in $\mathrm{Sim}(G_0)$ *at $\rho(s)$.*

Proof. First, for $m \in M(G_1)$, $E_{G_1}(s, m)$ is defined if and only if the pair move $E_{G_1}(s, m, 0)$ (in the order of the pair prescribed by the grading of m) is defined. Indeed, the latter presupposes the former, and conversely if $E_{G_1}(s, m)$ is defined then so is $E_{G_1}(s, 0)$, as s is not final, and so by the properties of Definition 4.1 the pair move is defined, using that the pseudo pass move has both gradings. If some $a \in \rho(s)$ were final, then no joint move would be defined in $\mathrm{Sim}(G_0)$ at $\rho(s)$, by the construction of $\mathrm{Sim}(G_0)$; by condition 3 no joint move would then be defined at s, contradicting the definedness of the pair $(0, 0)$ at the non final s. So every branch is non final. Now if $m \in M(s)$, the pair move is defined at s, hence, by condition 3, defined in $\mathrm{Sim}(G_0)$ at $\rho(s)$, which says exactly that $a\#m$ and $a\#0$ are defined for every $a \in \rho(s)$, that is $m \in \mathbb{M}(a)$ for every a. The equality statement is the same unwinding read in both directions. □

Definition 4.18. Let G_1 be a simultaneization of G_0. We say that G_1 is ***faithful*** at a state $s \in B(G_1)$ if $\rho(s)$ is nonempty, s is final only if some branch of $\rho(s)$ is final, and, when s is not final,

$$M(s) = \bigcap_{a \in \rho(s)} \mathbb{M}(a),$$

that is, exactly the joint moves of $\mathrm{Sim}(G_0)$ at $\rho(s)$ are available at s. The finality clause says that G_1 may not invent endings where it is faithful. For $d \geq 0$, we say that G_1 is ***faithful within distance*** d ***of classical play*** if it is faithful at every state reachable from a semi-classical state in at most d turns, distance 0 being the semi-classical states themselves.

Remark 4.19. *Faithfulness everywhere would say that G_1 is* $\mathrm{Sim}(G_0)$ *itself, up to the pruning of condition 3. Faithfulness near classical play says that the branch reduction is exact where the play is classical, and unconstrained beyond. The latter is what the decoherent variant of Section 5.1 achieves; the distinction carries the content of Theorem 5.7 below.*

Definition 4.20. Let G_0 be a sequential game. We will say that a simultaneization G_1 of G_0 is a ***simple simultaneization of*** G_0 if for every non final $s \in B(G_1)$ at which a non-pass move is available there is a legally obtainable position $a \in S(G_0)$ with

$$\mathbb{M}(a) = M(s),$$

where for s the initial state of G_1, a can be taken to be q_0.

Remark 4.21. *Equivalently, there is a map ϕ to $S(G_0)$, defined on the states of $B(G_1)$ in question, taking the initial state of G_1 to q_0, s.t. for all such s, $m \in M(G_0)$: $E_{G_1}(s, m)$ is defined if and only if $\phi(s)\#m$ is defined and s is not final. Note that no compatibility of ϕ with the game evolutions is asked for.*

By Lemma 4.17, a state s is semantically the set of branches $\rho(s)$. Simplicity says that, nevertheless, the moves available at s are always the moves of a single classical position of G_0. So the game can be refereed classically. The dynamical content of being a simultaneization is carried by ρ, at the level of branches. This is a natural division. Requiring ϕ to also intertwine the evolutions would be too strong: no single classical position can track the captures of SGo, so even SGo would fail to be simple. There may certainly be other reasonable notions of "simplicity".

5. Main theorems

5.1. **The decoherent variant** ***DSGo*****.** We now introduce a theoretical variant *DSGo*: decoherence-postprocessed *SGo*. Each turn is resolved by the rules of *SGo* – the objective reduction runs first, its captures standing – and the resulting state is then postprocessed as follows. The displayed board is measured against the branch evolution of the turn, nothing being uncaptured or replayed on the board itself. A state of *DSGo* carries the same data as one of *SGo* – a displayed board together with its entanglement state $\rho(s)$ – and moves are available as in *SGo*.

The serialization map extends to sets of positions elementwise,

$$P(\mathfrak{B}, m_0, m_1) = \bigcup_{a \in \mathfrak{B}} P(a, m_0, m_1);$$

we call $P(\rho(s), m_0, m_1)$ the *decoherence* of the turn m_0/m_1 at s. Next, the *postprocessing* $\Delta(s', \mathfrak{M})$ of a state s' by a finite set $\mathfrak{M}$ of positions is defined as follows: on the displayed board of s', a stone whose intersection is occupied in no element of $\mathfrak{M}$ is removed, and a q-stone created on this turn carrying a single color over the elements of $\mathfrak{M}$ becomes that color; the entanglement state is then re-cut to the elements of $\mathfrak{M}$ occupying no empty intersection of the postprocessed board. The evolution of *DSGo* is the postprocessed evolution of *SGo*:

$$E_{DSGo}(s, m_0, m_1) = \Delta\big(E_{SGo}(s, m_0, m_1),\ P(\rho(s), m_0, m_1)\big).$$

Remark 5.1. *The decoherence above is of the* first order*. Decoherence of order d – decohering the last d turns – would push the faithfulness of Theorem 5.2 to distance d, at an exponential price in bookkeeping. The $d \to \infty$ limit is exactly* $\mathrm{Sim}(Go)$*. In this language DSGo is SGo plus order one decoherence. Order zero – SGo itself – does not suffice: SGo is not faithful already at distance one, as the 3×3 exhaustion in the proof of Theorem 5.2 verifies.*

Let *Go* be the classical game with Chinese style area scoring and standard pass move, with the following stipulations: suicide is a legal move, as for example in the Tromp–Taylor rules [24, 23], there is no Ko or superko rule, and the game is ended analogously to the way *SGo* is ended, so that the role of a superko rule is played by the end of the game rules.

Theorem 5.2. *SGo and DSGo are simple simultaneizations of Go, and DSGo is faithful within distance one of classical play.*

Proof. The first two properties are by construction and Lemma 4.3.

Let $\rho : B(SGo) \to B(\mathrm{Sim}(Go))$ send a state to its entanglement component: the entanglement state is a finite set of positions of *Go*, that is an element of $B(\mathrm{Sim}(Go))$ (positions of *Go* being identified with their pass classes, via the grading normalization in the definition of P). Then $\rho(q_0) = \{q_0\} = q_0'$. The branch evolution of Section 2.1 is precisely the evolution by the serialization map P: playing the two moves in both orders, with a move to an occupied intersection forfeited, is the prescription defining P. So $E_{SGo}(s, m_0, m_1)$ is obtained by applying the resolution prunings to $E_{\mathrm{Sim}(Go)}(\rho(s), m_0, m_1)$, and the inclusion in condition 3 holds by construction. For the definedness direction of condition 3: a move of SGo is available if and only if its intersection is empty on the displayed board; a branch kept by the reduction has all its stones within the displayed board, a branch occupying a displayed-empty intersection being incompatible with the display and discarded; so a displayed-empty intersection is empty in every branch, and the joint move is available in $\mathrm{Sim}(Go)$ at $\rho(s)$.

For the semi-classicality condition: along a generating history of Definition 4.15 every turn commutes, so each branch evolution produces a single position, and the entanglement state remains a singleton; the prunings preserve it, since at these turns the two serializations agree, and a stone is removed only if it is removed in this common serialization; so $\rho(s)$ is the required singleton. Condition 5 holds since whenever both composites are defined, the moves did not interfere and the composites agree.

To verify the simplicity property: since in our version of Go suicide is a legal move and there is no Ko rule, for any non final position a of Go the interface $\mathbb{M}(a)$ consists of the pass moves and the moves of either player to the empty intersections of a. Likewise, for a non final $s \in B(SGo)$, $M(s)$ consists of the pass moves and the moves of either player to the empty intersections of s. So it suffices to exhibit a legally obtainable position $\phi(s)$ of Go with the same empty intersections as s. Let $\phi(s)$ be obtained by replacing every stone of s by a black stone. To see that $\phi(s)$ is legally obtainable, fix an empty intersection e of s, which exists as a non-pass move is available at s, and let Black play the stones of $\phi(s)$ in order of weakly decreasing distance to e, while White always passes. When a stone is placed, a geodesic from it to e exits the set of already placed stones at an empty intersection adjacent to the set, so at every moment every black component has a liberty, and no captures or suicides occur.

The preceding applies verbatim to both games. Finally, for the faithfulness clause of $DSGo$. We first formalize final states of $DSGo$ as follows. A state s of $DSGo$ is final exactly when some branch of $\rho(s)$ is final (of course this requires fixing a formalization of final states of Go). Then the finality clause holds by definition.

For the move clause, let s be semi-classical with associated position a, so that $\rho(s) = \{a\}$ by the semi-classicality condition, verified above. First,

$$M(s) = \mathbb{M}(a).$$

Indeed $M(s) \subseteq \mathbb{M}(a)$ is Lemma 4.17. Conversely, if s is not the initial state then $s = E_{DSGo}(s_-, m, m')$ for a semi-classical s_- with position a_- and a pair commuting at a_- (Definition 4.15); the decoherence of that turn is $P(\{a_-\}, m, m') = \{a\}$, a singleton, so by the postprocessing every stone of the displayed board of s is occupied in a: the displayed board has exactly the empty intersections of a, and $\mathbb{M}(a) \subseteq M(s)$. (For s the initial state both boards are empty.)

Now let $s' = E_{DSGo}(s, m_0, m_1)$ be any state one turn from s. By condition 3,

$$\rho(s') \subseteq E_{\mathrm{Sim}(Go)}(\{a\}, m_0, m_1) = P(a, m_0, m_1),$$

which is exactly the decoherence $P(\rho(s), m_0, m_1)$ of the turn. Suppose that some element b_0 of $P(a, m_0, m_1)$ occupies no intersection emptied by the SGo resolution of the turn: every intersection the resolution empties is empty in b_0. Then b_0 is discarded neither by the resolution nor by the re-cut, since a stone removed by the postprocessing is occupied in no element at all; so $\rho(s') \neq \emptyset$. Suppose moreover that every stone of the postprocessed board is occupied in some element of $\rho(s')$. Then an intersection is empty on the displayed board of s' exactly when it is empty in every element of $\rho(s')$, that is $M(s') = \bigcap_{b \in \rho(s')} \mathbb{M}(b)$: faithfulness at s'.

The suppositions hold by construction. At a semi-classical state the capture resolution of *SGo* is designed exactly so that some one serialization of the turn performs every capture of the resolution, and so that the resolution retains no stone that every surviving serialization captures.[10]

A complete formalization of the semi-classical case would amount to re-deriving the capture conditions of Section 2 in the automaton formalism, which we do not undertake.) On the 3×3 board the theorem is computer verified in total by exhaustion. □

It is natural to ask how much of the entanglement state the board itself determines:

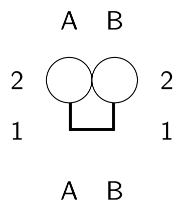
Question 5.3. *To what extent is the entanglement state of a legal SGo history determined by its sequence of displayed boards? We do not know. Example 5.6 below tempers expectations: branches interact through the display even in the absence of q-stones. In our samples about one in nine quantum-free displayed boards carried an entanglement state of more than one branch. Yet in all our computer experiments, whenever two legal histories arrived at the same displayed board with the same time stamps, their entanglement states were comparable: one was contained in the other. (About a hundred thousand sampled coincidences of displayed board and time stamps, on the 4×4 board.) So the display does not determine the entanglement state. But every divergence we have observed is of the mildest kind. The smaller of the two states is obtained from the larger by a branch reduction, and so is consistent with either history.*

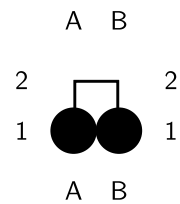
The faithfulness clause of Theorem 5.2 is not a formality. A faithful simultaneization exists for free – $\mathrm{Sim}(Go)$ itself, faithful everywhere by definition – but it fails simplicity read strictly (Remark 5.5 below), and as a game it degenerates: its dynamics can die within two turns.

Proposition 5.4. *On the 2×2 board, two turns of play can bring $\mathrm{Sim}(Go)$ to a state at which only the pass moves are ever available again. In exhaustive enumeration, 60% of the reachable states of $\mathrm{Sim}(Go)$ on the 2×2 board are so locked.*

Proof. Play turn one $A1/A2$ and turn two $B1/B2$. Turn two is a mutual capture race – each order of play, the stone landing second captures the other's pair – so the resulting state is the pair of branches:

[10] "Some" is exact. In the first example of Section 3.1 the resolution performs the captures of one serialization and discards the other reading: "every" fails. In a mutual capture race, as in the proof of Proposition 5.4, each serialization captures a side and the resolution captures neither: the converse fails, by design. At entangled states the single witness itself fails, and Example 5.6 is built on its failure: each of its two captures is performed by some reading, but no single reading performs both, and the entanglement empties.



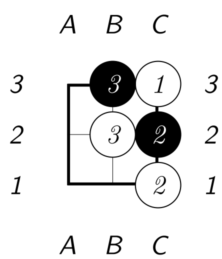


The two footprints together cover the board, so no stone move is available in Sim(Go), and passing changes nothing. □

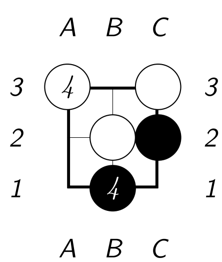
Remark 5.5. *This quantifies requirement (4) of Section 2.2. It also shows that* Sim(Go) *fails the simplicity condition at such states read strictly: no classical position has the pass-only interface, every legal position having an empty intersection. This is why Definition 4.20 exempts pass-locked states; in SGo the display protocol of Section 2.7 ends the corresponding states instead.*

In Sim(G_0) the branches never interact: the evolution is the union of independent classical evolutions. In *SGo* they do interact, through the displayed board; we might call these quantum effects.

Example 5.6 (Interference)**.** *On the* 3×3 *board play the history pass/*$C3$*;* $C2/C1$*;* $B3/B2$*, arriving at the displayed board:*



Both C3 and C2 are trapped yet uncaptured. In one reading of the entanglement Black's B3 has captured C3; in the other White's B2 has captured C2. This is a standoff on pure black and white stones, with no q-stone anywhere in the history. Now play the joint move $B1/A3$*. The displayed resolution captures both B3 and C1:*



```
8 | .  n  b  q  k  b  n  r
7 | .  p  p  p  p  p  p  p
6 | .  r  .  .  .  .  .  .
5 | p  .  .  .  .  .  .  .
4 | P  .  .  .  .  .  .  .
3 | .  R  .  .  .  .  .  .
2 | .  P  P  P  P  P  P  P
1 | .  N  B  Q  K  B  N  R
    a  b  c  d  e  f  g  h
```

FIGURE 1. The position a^*, reached by the mirror pairs $a4/a5$; $Ra3/Ra6$; $Rb3/Rb6$. White pieces are in capitals. The conflicting simultaneous moves are the rook moves $m_0 : b3 \to b5$ and $m_1 = \mathcal{A}(m_0) : b6 \to b4$.

though each single reading captures only one of them. The other survives on a liberty freed by that reading's earlier capture. The display remembers both stones of the standoff. It thus performs a joint capture that no serialization of the history performs. Every reading is then incompatible with the display, and the entanglement reduces to the empty set. Effects of this kind are what the faithfulness clause of Theorem 5.2 excludes from DSGo near classical play.

5.2. **Chess and faithfulness.** We now come to the main reason for studying faithfulness: it gives rise to a qualitative mathematical difference between chess and *Go*.

Theorem 5.7. *$\widetilde{chess}$ has no simple simultaneization that is faithful within distance one of classical play. In particular, the universal simultaneization* $\mathrm{Sim}(\widetilde{chess})$ *is not simple.*

Proof. First a technical lemma.

Lemma 5.8. *Let G_1 be a simultaneization of G_0, and suppose that $s \in B(G_1)$ is obtained from the initial state by a game history consisting of* mirror pairs*, that is, of simultaneous pairs of the form $(x_i, \mathcal{A}(x_i))$. Then $\mathcal{R}(s) = s$, and consequently $\mathcal{A}(M(s)) = M(s)$.*

Proof. The mirror of the pair $(x, \mathcal{A}(x))$ is the pair $(\mathcal{A}(\mathcal{A}(x)), \mathcal{A}(x)) = (x, \mathcal{A}(x))$ itself. So applying the symmetry property of Definition 4.2, and the order invariance of E (the last property of Definition 4.1), along the game history, together with $\mathcal{R}(q_0) = q_0$, we get $\mathcal{R}(s) = s$. Then for any m, $E_{G_1}(s, m)$ is defined if and only if $E_{G_1}(\mathcal{R}(s), \mathcal{A}(m)) = E_{G_1}(s, \mathcal{A}(m))$ is defined, giving $\mathcal{A}(M(s)) = M(s)$. $\square$

Now return to the proof of the theorem. Let G_1 be any simultaneization of $\widetilde{chess}$ faithful within distance one of classical play, with P_0 the White player, and consider the game history of G_1 consisting of the mirror pairs

$$a4/a5;\quad Ra3/Ra6;\quad Rb3/Rb6,$$

in standard chess algebraic notation. The moves of each pair do not interact, so both serializations are pseudo legal with the same outcome, and each state along this history is semi-classical. By the semi-classicality condition of Definition 4.16

together with faithfulness at the semi-classical states, applied inductively along the history, the history is legal in G_1 and for the resulting state $s^* \in B(G_1)$ we have $\rho(s^*) = \{a^*\}$, with a^* the classical position of Figure 1. Note that this pins down $\rho(s^*)$ for every simultaneization G_1, whatever its branch reduction mechanism.

Now let m_0 be the White rook move $b3 \to b5$, and $m_1 = \mathcal{A}(m_0)$ the Black rook move $b6 \to b4$. Each order of these moves blocks the other: after $b3 \to b5$ the Black rook's path is blocked at $b5$, while after $b6 \to b4$ the White rook's path is blocked at $b4$. So in $\mathrm{Sim}(\widetilde{chess})$ this pair resolves by the lowest priority case of the serialization map P, and

$$P(a^*, m_0, m_1) = \{a_0, a_1\},$$

where in a_0 the White rook stands on $b5$ with the Black rook on $b6$, and a_1 is the mirror image of a_0. By faithfulness at the semi-classical s^*, the pair (m_0, m_1), being available in $\mathrm{Sim}(\widetilde{chess})$ at $\{a^*\}$, is available at s^*; so $s_1^* = E_{G_1}(s^*, m_0, m_1)$ is defined, and by condition 3 together with faithfulness at s_1^* – a state one turn from s^* – $\rho(s_1^*)$ is a non-empty subset of $\{a_0, a_1\}$. Moreover s_1^* is not final: it is within distance one of the semi-classical s^*, so the finality clause of faithfulness applies, and neither a_0 nor a_1 is final in $\widetilde{chess}$.

Suppose that $\rho(s_1^*)$ is a singleton, say $\{a_0\}$. The history of s_1^* consists of mirror pairs, so by Lemma 5.8 $M(s_1^*)$ is $\mathcal{A}$ invariant. But by faithfulness at s_1^*, $M(s_1^*) = \mathbb{M}(a_0)$, which is not $\mathcal{A}$ invariant: in a_0 White has rook moves from $b5$, while Black has no rook moves from $b4$. The case of $\{a_1\}$ is mirrored. Thus this branch reduction is forbidden by the symmetry of G_1, and $\rho(s_1^*) = \{a_0, a_1\}$.

By faithfulness at s_1^*, $M(s_1^*) = \mathbb{M}(a_0) \cap \mathbb{M}(a_1)$, which is computed directly: for each player it consists of the pass move, the moves of the c through h pawns, and the four knight moves. There are no queen side rook moves, as these rooks stand on different squares in a_0 and a_1, and no a or b pawn moves. Now suppose that G_1 is simple, so that there is a legally obtainable position c of $\widetilde{chess}$ with $\mathbb{M}(c)$ equal to this set. The interface $\mathbb{M}(c)$ forces every piece of c other than the two queen side rooks, the two a pawns and the two b pawns to stand on its starting square. One now checks the finitely many placements, or absences by capture, of the six remaining pieces, and each fails. For example, if the White b pawn is at $b2$, then $b3$ must be occupied, necessarily by a queen side rook or the Black b pawn, giving respectively a legal rook move from $b3$ and the legal capture $b3 \times c2$, neither of which is in the set; while if the White b pawn is absent, having been captured, and $b2$ is unoccupied, then the $c1$ bishop has the legal move to $b2$, again not in the set. So we have a contradiction, and G_1 is not simple. □

5.3. **Mirror-tie chess.** The faithfulness hypothesis in Theorem 5.7 cannot be dropped. Define the *mirror-tie game* G_1: play proceeds as in $\widetilde{chess}$ but with simultaneously declared moves, as long as the two moves of each turn do not genuinely interact – both orders of play legal and with the same outcome. Thus each state s so reached is semi-classical, with $\rho(s)$ the singleton of its classical position. At the first genuinely conflicting move pair the game locks: only the pass moves remain available. After two consecutive passes the game ends in a tie.

The conditions of Definition 4.16 are readily verified; the locked states, having only passes available, are exempt from the simplicity condition, and at every other state the interface is that of a classical position. So G_1 is a simple simultaneization of $\widetilde{chess}$, faithful at distance zero. (Fully degenerate examples also exist: the game in which only passes are ever available is a simple simultaneization of any G_0 with a pass move, vacuously.) Together with Theorem 5.7: the faithfulness radius of $\widetilde{chess}$, over simple simultaneizations, is exactly zero – its conflicts can be refused, but not played out. Theorem 5.2 places *Go* strictly above this: the decoherent variant *DSGo* plays the conflicts out to radius at least one; how far the radius of *Go* extends we do not know.

6. Informed play and concluding remarks

The focus of this section is our informed play experiments: their methodology and partial results, for both games. All figures below are for the 5×5 board, and may be compared directly with the random play columns of the table of Section 2.2.

6.1. **Equilibrium guided play.** Our equilibrium guided play is computed as follows: at each turn both players are offered a random menu of five moves; the resulting 5×5 matrix of joint outcomes is filled by averaging the final margins of short uniformly random continuations;[11] and each player samples their move from an optimal mixed strategy of this zero sum matrix game. Three technical findings. First, at about 74% of the turns the estimated one turn game has *no pure saddle point*, so the players must randomize – consistent with Section 4.1, where mixing is proved necessary already on the 3×3 board (re-estimating the matrices at fivefold rollout depth, on fixed positions and menus, lowers the mixing fraction from 74% to 54% on a control sample; genuine mixing remains). Second, the within turn value spread is large, about 16 points of the twenty five: the joint move choice of a single turn swings most of the board. Third, the summary table of the next subsection shows that equilibrium guided play leaves the creation of entanglement essentially unchanged relative to the random play columns of the table of Section 2.2 (3.2 q-stones created per game against 3.3) and resolves slightly more of it by the end (final branch count 3.2 against 3.4): the entanglement is a stable feature of the game, of order one under random and informed play alike.

6.2. **Versus informed play in classical *Go*.** For the classical game we use greedy rollout guided play: at each move the player evaluates up to six candidate moves, and the pass, by twelve random rollouts each (the convention of the preceding footnote), and plays the best. The summary of informed self play, for both games, in the format of the table of Section 2.2 (162 and 165 games; unbounded play in both cases):

[11] Rollouts use the standard Monte Carlo playout convention: uniformly random moves, except that a player does not fill their own single point eyes and does not play suicide, passing when no other move is available.

	SGo	classical *Go*
mean margin	11.7	11.4
draws	5.6%	6.7%
games ending on their own	100%	100%
mean length in turns	21.2	17.8
q-stones created per game	3.2	–
q-stones on the final board	2.3	–
peak number of branches	4.2	–
final number of branches	3.2	–

For *SGo* the table is close to its random play columns: the margin eases (11.7 against 13.6), the draw rate and game length barely move, and the entanglement is unchanged in creation and slightly more resolved by the end. For the classical game the change is termination: informed play supplies the endings that random play never reaches – every game now ends on its own, in 17.8 turns on average – while *SGo* terminates by rule either way.

At this shallow search depth no measurable first mover advantage emerges in *Go* (signed mean margins statistically zero). Hence we are not correcting by any Komi, which is also why draws appear for *Go* in the table above. We note that *Go* is solved on the 5×5 board – a win for Black by 25 points under standard rule formalizations [25] – a reminder that these are partial, shallow depth results.

Whether the above observations persist for equilibrium guided play on large boards, along with the sharpness observed in Section 2.2, and to what extent the visible board determines the entanglement state (Question 5.3), we leave to future study, which may well require the assistance of higher level compute resources.

Finally, we note that the pattern of resolution in *SGo* – branchings accumulate silently until objective information forces their evaluation – is also familiar in computer science, notably in lazy evaluation [8, 11], where a deferred computation is forced exactly when its value is demanded; we thank a referee for this observation.

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